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The Editor
American Journal of Physics

Dear Editor,

We submit “How long is a simulated double pendulum right? Step size, floating-point precision, and the Lyapunov exponent” for consideration in the Computational Physics section.

The manuscript is a teaching contribution and presents no new physics. It describes a computational exercise for upper-division courses in which students integrate the double pendulum in arbitrary-precision arithmetic, vary the step and the precision independently, and find that the time for which the simulation is accurate obeys one logarithmic law governed by the largest Lyapunov exponent. Students measure that exponent from their own numerical error and compare it with the standard two-trajectory estimate. Along the way they meet a best step for each floating-point format, a ceiling on what double precision can achieve, and a trajectory that is entirely wrong while conserving energy to many digits. The manuscript ends with four suggested problems.

The Journal has treated the double pendulum as a vehicle for teaching chaos since Shinbrot *et al.* [60, 491 (1992)] and Levien and Tan [61, 1038 (1993)]. Those papers concern the physical system and how it amplifies an uncertainty in its initial state. Ours concerns the simulation that instructors and students now routinely substitute for the apparatus, and asks how long it can be believed. We are not aware of an undergraduate-level treatment in which students separate truncation from roundoff, measure each, and connect both to the Lyapunov exponent. By the Journal’s rule of thumb the manuscript is about five journal pages, and a two-column build is just over six with the references.

In accordance with the Journal’s procedure the manuscript file contains no author names, affiliations or acknowledgments. The supplementary material comprises the source code, the plotted data and two animations. The work is original and is not under consideration elsewhere, and both authors have approved its submission.

Sincerely,

Arjun Pemmasani
on behalf of both authors